

Foolproof Weak Roman Domination Problem *

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Abstract

Given a graph G , a guard function $f : V \rightarrow \{0, 1, 2\}$ of G decomposes the vertex set into V_0, V_1 , and V_2 such that $V_i = \{v | f(v) = i\}$ for $0 \leq i \leq 2$, and for each $v \in V_0$, $N(v) \cap (V_1 \cup V_2) \neq \emptyset$ where $N(v) = \{u | uv \in E\}$. For a vertex $v \in V_0$, and $u \in N(v) \cap (V_1 \cup V_2)$, a move function $g(f : u \rightarrow v)$ respected to a guard function f is defined as $g(v) = 1$, $g(u) = f(u) - 1$, and $g(w) = f(w)$ for $w \in V - \{u, v\}$. A foolproof weak Roman dominating function (FWRDF) f of a graph G is a guard function such that for each $v \in V_0$, every $u \in N(v) \cap (V_1 \cup V_2)$, the move function $g(f : u \rightarrow v)$ is a guard function of G . The weight of a FWRDF is defined as $w(f) = \sum_{v \in V} f(v) = 2|V_2| + |V_1|$. The foolproof weak Roman domination number is the minimum weight of a FWRDF among all FWRDFs of G , denoted as $r_r^*(G)$. The foolproof weak Roman domination problem is to find the foolproof weak Roman domination number of any graph G . This paper established the upper bound of foolproof weak Roman domination number on the spider web graph.

1 Introduction

Roman domination problem was derived from the Roman legions deploy in the fourth century. This problem can be modeled by a graph where the vertices represent the regions and there is an edge between two vertices if the corresponding regions are adjacent to each other. Any region without a legion stationed must neighbor another region having two legions stationed. If the further were attacked, then the later could deploy a legion to rescue it without becoming vulnerable itself. For a simple graph $G = (V, E)$, a Roman domination function (RDF) $f : V(G) \rightarrow \{0, 1, 2\}$ satisfies that for every vertex v with $f(v) = 0$ is adjacent to a vertex u such that $f(u) = 2$. A Roman domination function corresponds to the deployment of legions in each region in the whole empire. The weight

(or cost) of a RDF f is $w(f) = \sum_{v \in V} f(v) = 2|V_2| + |V_1|$. The Roman domination number, defined as $r_R(G) = \min\{w(f) | f \text{ is a RDF}\}$, the Roman domination number refers to the minimum legions needed to protect the whole empire. In 2003, Henning and Hedetniemi proposed the weak Roman domination in order to further reduce the number of legions to protect the empire. This version allows the neighboring region to rescue each other even if they have only one legion as long as after the deployment, every region without a legion stationed is still adjacent to a region having legion stationed. Let $f : V(G) \rightarrow \{0, 1, 2\}$ be a function of G and $V_i = \{u | f(u) = i\}$ for $0 \leq i \leq 2$. Then f is a guard function if it has the property that every vertex $v \in V_0$ is adjacent to a vertex $u \notin V_0$. For each $v \in V_0$, define a move function $g(f : u \rightarrow v)$ respected to function f as

$$g(w) = \begin{cases} 1, & \text{if } w = v; \\ f(u) - 1, & \text{if } w = u; \\ f(w), & \text{otherwise.} \end{cases}$$

Then a guard function $f : V(G) \rightarrow \{0, 1, 2\}$ is a weak Roman dominating function (WRDF) if and only if for every $v \in V_0$, there is a move function g which results a guard function of G . The weight (or cost) of a WRDF is $w(f) = \sum_{v \in V} f(v) = 2|V_2| + |V_1|$. The weak Roman domination number, denoted as $r_r(G)$, is the minimum weight of the WRDF among all WRDFs of G .

Weak Roman domination problem has been proved to be NP-Complete even if the graph is restricted to bipartite or chordal graphs [7]. The polynomial solution to this problem is found for a limited classes of graphs such as complete graph, path, cycle and complete multipartite graph [4][7][9]. The upper bound of $r_r(G)$ is known for some complex graph structures, such as grid graph, cartesian product of cycle with cycle, complete graph with complete graph and claw-free graph with claw-free graph [3][4]. Some are interested in protecting the graph with finite moves [1][6] others are interested in protecting the graph with unlimited moves [2][5].

Since weak Roman domination requires some strategies to determine which move to make in or-

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der to resolve the problem of an attack on a vertex $v \in V_0$, is reasonable to consider foolproof versions of this dynamic configuration case where the moving strategy is not required. In 2004, the foolproof version was proposed by [1][2]. The neighborhood of a vertex v , denoted as $N(v)$, is the set of all adjacent vertices of v . The *close neighborhood* of v is defined as $N[v] = N(v) \cup \{v\}$. A *foolproof weak Roman dominating function* (FWRDF) is a guard function $f : V(G) \rightarrow \{0, 1, 2\}$, such that for every $v \in V_0$, the move function $g(f : u \rightarrow v)$ where $u \in (V_1 \cup V_2) \cap N(v)$ results a guard function of G . The minimum cost among all FWRDF on a given graph is denoted as $r_r^*(G)$.

Buger et al. investigated the FWRDF for a number of simple graphs such as path, cycle, complete bipartite graphs[1]. They also consider infinite sequence attacks[2] on path, cycle, complete multipartite graph, hexagonal graph, and cartesian product of complete graph with complete graph, cycle with cycle, and path with path.

This paper considers the foolproof weak Roman domination problem on a generalized spider web graph $SW(m, n)$.

2 Preliminaries

A city is called protected if when it is attacked, it can be protected either by the legion stationed in it or by the legion send from its neighbor.

A spider web graph (defined in [8]) is denoted as $SW(m, n)$ where m and n are even integers with $m \geq 4, n \geq 2$. $SW(m, n)$ has vertex set $\{(i, j) | 0 \leq i < m, 0 \leq j < n\}$ such that (i, j) and (k, l) are adjacent if they satisfy one of the following conditions:

1. $i = k$ and $l = j + 1$ for $0 \leq j \leq n - 2$
2. $j = l$ and $k = [i + 1]_m$ if $i + j$ is even
3. $j = l$ and $k = [i + 1]_m$ if $j = n - 1$ or $j = 0$

where $[r]_s$ denote $r \pmod s$. For example, a spider graph $SW(8, 6)$ is illustrated in Figure 1.

This paper considers the foolproof weak Roman domination problem on a generalized spider web graph $SW(m, n)$ for $m \geq 4, n \geq 2$ which only require m is even.

Since there are n layers of the $SW(m, n)$, for convenience let the inner cycle $C_m^0 = v_{0,0}, v_{1,0}, \dots, v_{m-1,0}, v_{0,0}$ be the first cycle, and the outer cycle $C_m^{n-1} = v_{0,n-1}, v_{1,n-1}, \dots, v_{m-1,n-1}, v_{0,n-1}$ be the last cycle.

Lemma 1 Let $f : V(G) \rightarrow \{0, 1, 2\}$ be a guard function of G . Then f is a FWRDF of G if and only if for each $v \in V_0$, one of the following holds.

1. $\sum_{u \in N(v)} f(u) \geq 2$;

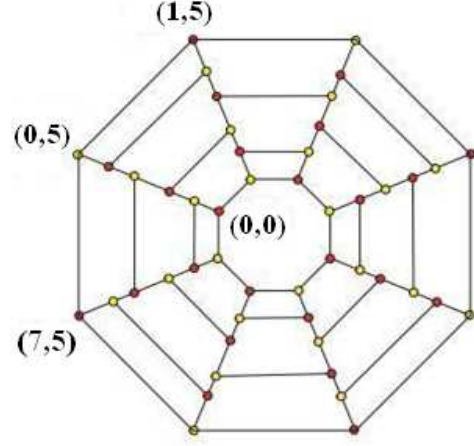


Figure 1: $SW(8, 6)$

2. $\sum_{u \in N(v)} f(u) = 1$ and $\{w\} = N(v) \cap V_1$ then $(N(w) - N[v]) \cap V_0 = \emptyset$;

Proof. First we show that the condition are necessary. Suppose f is a FWRDF of G , since $f(v) = 0$, by definition of FWRDF, $\sum_{u \in N(v)} f(u) \geq 1$. If $\sum f(u) \geq 2$, then condition(1) holds. So we may assume that $\sum_{u \in N(v)} f(u) = 1$. Suppose to the contrary that $(N(w) - N[v]) \cap V_0 \neq \emptyset$, then there exists a vertex p such that $p \in (N(w) - N[v]) \cap V_0$. Let w be the vertex to rescue p when p is attacked. After the movement, $f'(w) = 0, f'(p) = 1$ and $f'(x) = f(x)$ for $x \in V(G) - \{p, w\}$. Since $p \notin N[v]$, we know that $\sum_{u \in N(v)} f'(u) = 0$, which implies that f' is not a guard function of G contradicting to the fact that f is a FWRDF of G .

Next we show that the condition is sufficient. Let $g : V(G) \rightarrow \{0, 1, 2\}$ be a guard function satisfies the two conditions, we want to show that g is a FWRDF of G . When a vertex $v \in V_0$ is attacked, consider following cases:

Case 1. v is rescued by p , such that $g(p) = 2$. After the movement, $g'(v) = 1, g'(p) = 1$ and $g'(x) = g(x)$ for $x \in V(G) - \{p, v\}$. Then $\forall w \in N(p)$, $\sum_{u \in N(w)} g'(u) \geq 1$, which implies that g' is also a guard function of G .

Case 2. v is rescued by t , such that $g(t) = 1$. Then $g'(v) = 1, g'(t) = 0$ and $g'(x) = g(x)$ for $x \in V(G) - \{t, v\}$. Let $q \in N(t) - \{v\}$ and $g(q) = 0$, if $q \in N(v)$, then $\sum_{u \in N(q)} g'(u) \geq 1$, if $q \notin N(v)$, then g must satisfies condition(1), which implies $\sum_{u \in N(q)} g(u) \geq 2$, that is $\sum_{u \in N(q)} g'(u) \geq 1$. Hence g is a FWRDF of G . ■

3 Experiments

Theorem 1 Let $G = (V, E) = SW(m, 2t)$ for $1 \leq t \leq 2$. Then $r_r^*(G) \leq tm$.

Proof. We prove the theorem by construct a FWRDF of G with cost tm . Consider function $f : V \rightarrow \{0, 1, 2\}$ which decomposes V into V_0 , V_1 , and V_2 such that $V_2 = \emptyset$, $V_1 = \{v_{i,j} | [i+j]_2 = 0\}$, and $V_0 = V - V_1$. Note that if $v_{i,j} \in V_0$ then $v_{i,j+1}, v_{i,j-1}, v_{i+1,j}$, and $v_{i-1,j}$ are in V_1 . Hence for each $v \in V_0$, we have $\sum_{u \in N(v)} f(u) = 3$. By Lemma 1, f is a FWRDF of G with cost tm . ■

Figure 2 shows an example of the guard function of $SW(m, 4)$. The solid vertices denote the vertices in V_1 , and the hollow vertices without number denote the vertices in V_0 .

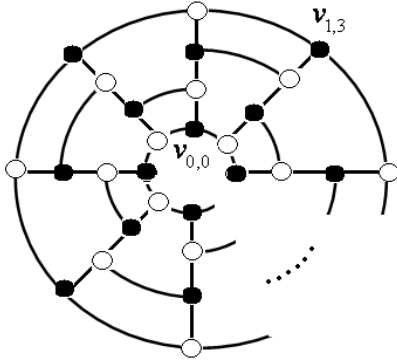


Figure 2: $SW(m, 4)$

Theorem 2 Let $G = (V, E) = SW(m, 3)$. Then $r_r^*(G) \leq m + \lceil m/3 \rceil$.

Proof. We prove the theorem by construct a FWRDF of G with cost $m + \lceil m/3 \rceil$. Consider function $f : V \rightarrow \{0, 1, 2\}$ where $V_2 = \emptyset$, $V_1 = \{v_{i,j} | j \in \{0, 2\} \text{ and } [i]_3 = 0; \text{ or } j = 1 \text{ and } [i] \in \{1, 2\}\}$, and $V_0 = V - V_1$. For any $v_{i,j} \in V_0$, if $[j]_3 = 0$, then $v_{i,j+1}$, and either $v_{i+1,j}$ or $v_{i-1,j}$ in V_1 ; else if $[j]_3 = 2$, then $v_{i,j-1}, v_{i,j+1}$, and either $v_{i+1,j}$ or $v_{i-1,j}$ in V_1 ; else if $[j]_3 = 1$, then $v_{i,j+1}, v_{i,j-1}$, and either $v_{i-1,j}$ or $v_{i+1,j}$ in V_1 ; $\sum f(v_{i,j}) = m$ for $0 \leq i \leq m-1, 0 \leq j \leq 1$, and $\sum f(v_{i,j}) = \lceil m/3 \rceil$ for $0 \leq i \leq m-1, j = 2$. By Lemma 1, f is a FWRDF of G with cost $m + \lceil m/3 \rceil$. ■

Figure 3 shows an example of the guard function of $SW(m, 4)$. The solid vertices denote the vertices in V_1 , and the hollow vertices without number denote the vertices in V_0 .

Next we consider case of $n \geq 5$ for $G = SW(m, n)$ where $[m]_4 = 0$. Algorithm 1 produces a guard function f of $SW(m, n)$. We show the guard function is a FWRDF and the cost of the function is no more than $m(2n - \lfloor (n-1)/4 \rfloor)/4$. Algorithm 1 consider $f(v_{i,j})$ layer by layer from the inner cycle to the

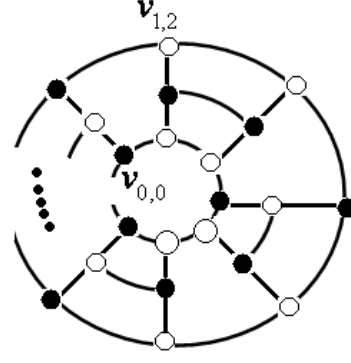


Figure 3: $SW(m, 3)$

outer cycle which will loop for each eight layers. Finally, when n is even, it adds extra cost on the outer cycle.

Algorithm 1 A guard function f of $SW(m, n)$ for $[m]_4 = 0$.

At first initialize $f(v) = 0$ for all $v \in SW(m, n)$

1. for $j = 0$ to $n - 1$
2. for $i = 0$ to $m - 1$
3. if $[j]_8 \in \{0, 4\}$
4. $f(v_{i,j}) = 1$ for even i
5. if $[j]_8 \in \{1, 3\}$
6. $f(v_{i,j}) = 1$ for $[i]_4 = 3$
7. if $[j]_8 = 2$
8. $f(v_{i,j}) = 2$ for $[i]_4 = 1$
9. $f(v_{i,j}) = 1$ for $[i]_4 = 2$
10. if $[j]_8 \in \{5, 7\}$
11. $f(v_{i,j}) = 1$ for $[i]_4 = 1$
12. if $[j]_8 = 6$
13. $f(v_{i,j}) = 2$ for $[i]_4 = 3$
14. $f(v_{i,j}) = 1$ for $[i]_4 = 0$
15. if $j = n - 1$ for even n
16. case 1. $[j]_8 = 1$
17. $f(v_{i,j}) = 1$ for $[i]_4 = 1$
18. case 2. $[j]_8 = 3$
19. $f(v_{i,j}) = 1$ for $[i]_4 = 0$
20. case 3. $[j]_8 = 5$
21. $f(v_{i,j}) = 1$ for $[i]_4 = 3$
22. case 4. $[j]_8 = 7$
23. $f(v_{i,j}) = 1$ for $[i]_4 = 2$

Theorem 3 Algorithm 1 produces a FWRDF of $SW(m, n)$ for $[m]_4 = 0$ and $n \geq 5$ with cost $m(2n - \lfloor (n-1)/4 \rfloor)/4$ in time complexity $O(mn)$.

Proof. Since algorithm 1 produce $f(v_{i,j})$ layer by layer from the inner cycle to the outer cycle which will loop for every eight layers, we may consider each $v_{i,j}$ according to $[i]_4$ in the following cases:

Case 1. $[i]_4 = 0$.

Let $h, s, 2$ denote the vertex $v_{i,j}$ such that $f(v_{i,j})$ is 0, 1, and 2 respectively. Then when $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{shhshshsh\}$ for each eight layers. We have $v_{i,j-1}, v_{i-1,j} \in V_1$ for the first 'h'; $v_{i+1,j} \in V_2$ for the second 'h'; $v_{i,j+1}, v_{i-1,j} \in V_1$ for the third 'h'; $v_{i,j-1}, v_{i,j+1} \in V_1$ for the fourth and fifth 'h'.

Case 2. $[i]_4 = 1$.

When $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{hh2hhshs\}$ for each eight layers. We have $v_{i+1,j}, v_{i-1,j} \in V_1$ for the first 'h'; $v_{i,j+1} \in V_2$ for the second 'h'; $v_{i,j-1} \in V_2$ for the third 'h'; $v_{i,j+1}, v_{i-1,j} \in V_1$ for the fourth 'h'. $v_{i,j+1}, v_{i,j-1} \in V_1$ for the fifth 'h'.

Case 3. $[i]_4 = 2$.

When $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{shshshhh\}$ for each eight layers. We have $v_{i,j+1}, v_{i,j-1} \in V_1$ for the first and second 'h'; $v_{i,j-1}, v_{i-1,j} \in V_1$ for the third 'h'; $v_{i+1,j} \in V_2$ for the fourth 'h'. $v_{i,j+1}, v_{i-1,j} \in V_1$ for the fifth 'h'.

Case 4. $[i]_4 = 3$.

When $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{hshshh2h\}$ for each eight layers. We have $v_{i,j+1}, v_{i-1,j}, v_{i+1,j} \in V_1$ for the first 'h'; $v_{i,j-1}, v_{i,j+1} \in V_1$ for the second 'h'; $v_{i,j-1}, v_{i-1,j} \in V_1$ for the third 'h'; $v_{i,j+1} \in V_2$ for the fourth 'h'. $v_{i,j-1} \in V_2$ for the fifth 'h'.

By the above cases we knew that for each $v \in V_0$, we have $\sum_{u \in N(v)} f(u) \geq 2$, by Lemma 1, the function produced by algorithm 1 is a FWRDF of $SW(m, n)$ for $[m]_4 = 0$ and $n \geq 5$.

If we sum up the cost in each layer we can see that for layer j such that $[j]_4 = 0$, $\sum f(v_{i,j}) = m/2$; $[j]_4 \in \{1, 3\}$, $\sum f(v_{i,j}) = m/4$; $[j]_4 = 2$, $\sum f(v_{i,j}) = 3m/4$. When n is even, the algorithm add extra cost $m/4$ to the outer cycle (see line 15-23). Hence, the cost of $SW(m, n)$ respect to f can be summed up as

$$w(f) = \begin{cases} k + m/2[n]_4, & \text{for } [n]_4 \in \{1, 2, 3\}; \\ k + m/4, & \text{for } [n]_4 = 0; \end{cases}$$

$$= m(2n - \lfloor (n-1)/4 \rfloor)/4.$$

where $k = \lfloor n/4 \rfloor (7m/4)$.

Since the algorithm has loops from 0 to $n-1$ then from 0 to $m-1$, the total time complexity is $O(mn)$. ■

Figure 4 shows an example of the guard function of $SW(4, 10)$. The solid vertices denote the vertices in V_1 , the hollow vertices without number denote the vertices in V_0 , and the vertices label 2 are represent the vertices in V_2 .

Next we consider case of $n \geq 5$ for $G = SW(m, n)$ where $[m]_4 = 2$. Algorithm 2 produces a guard function f of $SW(m, n)$. We show the guard function is a FWRDF and the cost of the function is no more

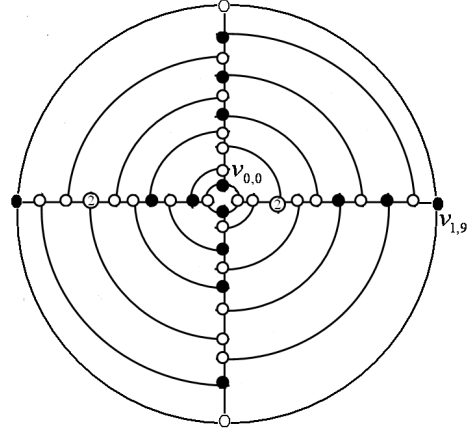


Figure 4: $SW(4, 10)$

than $(m-2)(2n - \lfloor (n-1)/4 \rfloor)/4 + n$. Algorithm 2 consider $f(v_{i,j})$ layer by layer from the inner cycle to the outer cycle which will loop for each eight layers. Finally, when n is even, it adds extra cost on the outer cycle.

Algorithm 2 A guard function f of $SW(m, n)$ for $[m]_4 = 2$.

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At first initialize  $f(v) = 0$  for all  $v \in SW(m, n)$ 
1. for  $j = 0$  to  $n-1$ 
2.   for  $i = 0$  to  $m-1$ 
3.     if  $[j]_8 \in \{0, 4\}$ 
4.        $f(v_{i,j}) = 1$  for even  $i$ 
5.     if  $[j]_8 \in \{1, 3\}$ 
6.        $f(v_{i,j}) = 1$  for  $[i]_4 = 3$  OR  $i = m-1$ 
7.     if  $[j]_8 = 2$ 
8.        $f(v_{i,j}) = 2$  for  $[i]_4 = 1$ 
9.        $f(v_{i,j}) = 1$  for  $[i]_4 = 2$  OR  $i = m-2$ 
10.    if  $[j]_8 \in \{5, 7\}$ 
11.       $f(v_{i,j}) = 1$  for  $[i]_4 = 1$  OR  $i = m-1$ 
12.    if  $[j]_8 = 6$ 
13.       $f(v_{i,j}) = 2$  for  $[i]_4 = 3$ 
14.       $f(v_{i,j}) = 1$  for  $[i]_4 = 0$  OR  $i = m-2$ 
15.    if  $j = n-1$  for even  $n$ 
16.      case 1.  $[j]_8 = 1$ 
17.         $f(v_{i,j}) = 1$  for  $[i]_4 = 1$ 
18.      case 2.  $[j]_8 = 3$ 
19.         $f(v_{i,j}) = 1$  for  $[i]_4 = 0$ 
20.      case 3.  $[j]_8 = 5$ 
21.         $f(v_{i,j}) = 1$  for  $[i]_4 = 3$ 
22.      case 4.  $[j]_8 = 7$ 
23.         $f(v_{i,j}) = 1$  for  $[i]_4 = 2$ 
    
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Theorem 4 Algorithm 2 produces a FWRDF of $SW(m, n)$ for $[m]_4 = 2$ and $n \geq 5$ with cost $(m-2)(2n - \lfloor (n-1)/4 \rfloor)/4 + n$ in time complexity $O(mn)$.

Proof. Since algorithm 2 produce $f(v_{i,j})$ layer by layer from the inner cycle to the outer cycle which

will loop for every eight layers, we may consider each $v_{i,j}$ according to $[i]_4$ in the following cases:

Case 1. $[i]_4 = 0$.

Let $h, s, 2$ denote the vertex $v_{i,j}$ such that $f(v_{i,j})$ is 0, 1, and 2 respectively. Then when $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{shhshshsh\}$ for each eight layers. We have $v_{i,j-1}, v_{i-1,j} \in V_1$ for the first 'h'; $v_{i+1,j} \in V_2$ for the second 'h'; $v_{i,j+1}, v_{i-1,j} \in V_1$ for the third 'h'; $v_{i,j-1}, v_{i,j+1} \in V_1$ for the fourth and fifth 'h'.

Case 2. $[i]_4 = 1$.

When $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{hh2hshshs\}$ for each eight layers. We have $v_{i+1,j}, v_{i-1,j} \in V_1$ for the first 'h'; $v_{i,j+1} \in V_2$ for the second 'h'; $v_{i,j-1} \in V_2$ for the third 'h'; $v_{i,j+1}, v_{i-1,j} \in V_1$ for the fourth 'h'. $v_{i,j+1}, v_{i,j-1} \in V_1$ for the fifth 'h'.

Case 3. $[i]_4 = 2$.

When $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{shshshhhh\}$ for each eight layers. We have $v_{i,j+1}, v_{i-1,j} \in V_1$ for the first and second 'h'; $v_{i,j-1}, v_{i-1,j} \in V_1$ for the third 'h'; $v_{i+1,j} \in V_2$ for the fourth 'h'. $v_{i,j+1}, v_{i-1,j} \in V_1$ for the fifth 'h'.

Case 4. $[i]_4 = 3$.

When $[i]_4 = 0$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{hshshsh2h\}$ for each eight layers. We have $v_{i,j+1}, v_{i-1,j}, v_{i+1,j} \in V_1$ for the first 'h'; $v_{i,j-1}, v_{i,j+1} \in V_1$ for the second 'h'; $v_{i,j-1}, v_{i-1,j} \in V_1$ for the third 'h'; $v_{i,j+1} \in V_2$ for the fourth 'h'. $v_{i,j-1} \in V_2$ for the fifth 'h'.

Case 5. $i = m - 2$.

When $i = m - 2$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{shshshshsh\}$ for each eight layers. We have $v_{i,j+1}, v_{i,j-1} \in V_1$ for all 'h'.

Case 6. $i = m - 1$.

When $i = m - 1$, the $v_{i,j}$ from inner cycle to outer cycle may be represented as $\{hshshshshs\}$ for each eight layers. We have $v_{i,j+1}, v_{i-1,j}, v_{i+1,j} \in V_1$ for the first 'h'; $v_{i,j+1}, v_{i,j-1} \in V_1$ for all the other 'h'.

By the above cases we knew that for each $v \in V_0$, we have $\sum_{u \in N(v)} f(u) \geq 2$, by Lemma 1, the function produced by algorithm 2 is a FWRDF of $SW(m, n)$ for $[m]_4 = 2$ and $n \geq 5$.

If we sum up the cost in each layer we can see that for layer j such that $[j]_4 = 0$, $\sum f(v_{i,j}) = 2(m-2)/4 + 1$; $[j]_4 \in \{1, 3\}$, $\sum f(v_{i,j}) = (m-2)/4 + 1$; $[j]_4 = 2$, $\sum f(v_{i,j}) = 3(m-2)/4 + 1$. When n is even, the algorithm add extra cost $(m-2)/4$ to the outer cycle (see line 15-23). Hence, the cost of $SW(m, n)$ respect to f can be summed up as

$$w(f) = \begin{cases} k + [n]_4(m-2)/2, & \text{for } [n]_4 \in \{1, 2, 3\}; \\ k + (m-2)/4, & \text{for } [n]_4 = 0; \end{cases}$$

$$= (m-2)(2n - \lfloor (n-1)/4 \rfloor)/4 + n.$$

where $k = \lfloor n/4 \rfloor (7(m-2)/4) + n$.

Since the algorithm has loops from 0 to $n-1$ then from 0 to $m-1$, the total time complexity is $O(mn)$. ■

Figure 5 shows an example of the guard function of $SW(6, 9)$. The solid vertices denote the vertices in V_1 , the hollow vertices without number denote the vertices in V_0 , and the vertices label 2 are represent the vertices in V_2 .

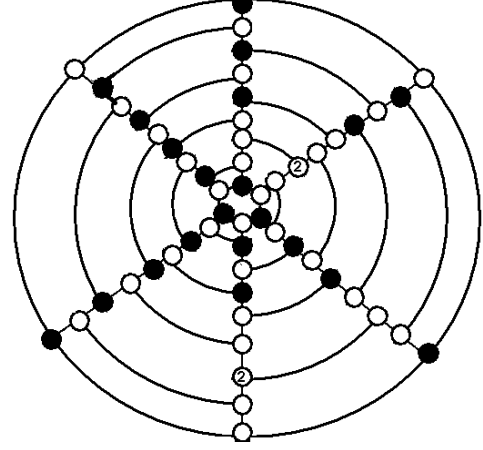


Figure 5: $SW(6, 9)$

4 Conclusion

In this paper, we provide the upper bound of the foolproof weak Roman domination number on a generalized spider web graph (shown in Table 1) in time complexity $O(mn)$.

Table 1: The upper bound of $r_r^*(G)$ on $SW(m, n)$.

n	The upper bound of $r_r^*(SW(m, n))$
$n = 2$ or 4	$mn/2$
$n = 3$	$m + \lceil m/3 \rceil$
$n \geq 5$	$m(2n - \lfloor (n-1)/4 \rfloor)/4$, if $[m]_4 = 0$
	$(m-2)(2n - \lfloor (n-1)/4 \rfloor)/4 + n$, if $[m]_4 = 2$

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